Pro Football Sports Science Research Update

December 2025

12-months MRI after anterior cruciate ligament reconstruction can identify risk factors for subsequent graft rupture and used to guide the return to sport – Results from a high-volume institution. Journal of ISAKOS, 14: 1000994, 2025.

Changes in eccentric hamstring strength and muscle soreness in younger and older male football players following match play. BMC Sports Science, Medicine and Rehabilitation, 17: 310, 2025.

Clenbuterol abuse in bodybuilding and athletics: Unsupervised use, psychological motivations, and health consequences. Cureus, 17(5): e84904, 2025.

Does maximal strength or speed-strength have stronger associations with performance indicators in the rugby football codes? A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-13, 2025.

Effect or oral and topical sodium bicarbonate on functional recovery and soccer-specific performance after exercise-induce muscle damage: A randomized, double-blind, placebo-controlled study. Nutrients, 17: 3383, 2025.

Effects of simulated short-haul flights on sport-specific performance, sleep, and psychophysiological responses in football players. Scandinavian Journal of Medicine & Science in Sports, 35: e70169, 2025.

Ensuring scientific rigor in research: Why sports-related publicly obtained data fall short. Foot & Ankle Orthopaedics, 10(2): 1, 2025.

Exploring the physiological mechanisms of sleep's influence on athletic performance and recovery: A narrative review. Sleep and Breathing, 29: 354, 2025.

Global research trends in sports nutrition and football over the last 20 years (2004-2024). Sports, 13: 365, 2025.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Gluteal muscle forces during hip-focused injury prevention and rehabilitation exercises. Medicine and Science in Sports and Exercise, 55(4): 650-660, 2025.

Letter to the Editor: Injectable therapeutic peptides – An adjunct to regenerative medicine and sports performance? Arthroscopy: The Journal of Arthroscopic and Related Surgery, 41(2): 150-151, 2025.

Intermediary role of mental toughness beliefs on the relationship between pain self-efficacy and fear avoidance in elite injured athletes. BMC Sports Science, Medicine and Rehabilitation, 17: 124, 2025.

Melatonin secretion and impacts of training and match schedules on sleep quality, recovery, and circadian rhythms in young professional football players. Biomolecules, 15: 700, 2025.

Neither American football playing position nor total years playing football are associated with neurobehavioral health outcomes: An NFL-LONG study. Current Sports Medicine Reports, 24(11): 372-380, 2025.

Physiological adaptations and practical efficacy of different blood flow restriction resistance training modes in athletic populations. Frontiers in Physiology, 16: 1683442, 2025.

Reciprocal relationships between sleep quality, mental health and the quality of life in elite athletes: A pilot study. Biology of Sport, 43: 267-279, 2026.

Should GPS data be normalized for performance and fatigue monitoring in soccer? A theoretical-practical discussion on high-speed running. Frontiers in Sports and Active Living, 7: 1603767, 2025.

Specificity and areas of usage of cardiovascular prediction models among athletes – State of the art review. Reviews in Cardiovascular Medicine, 26(5): 37493, 2025.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Speeding up or slowing down: The effect of decoupling speed on knee biomechanics and limb loading during split-belt treadmill training in persons with ACL reconstruction. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2025.

Terminology and interpretation across neuromuscular profiling methods: A systematic review. Sports Medicine, Ahead of Print: 1-31, 2025.

The association between central nervous system function and core body temperature in patients with exertional heat stroke. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2025.

The impact of weekly acceleration and deceleration loads on neuromuscular performance in soccer: A session-to-session analysis. Journal of Sports Science and Medicine, 24: 851-860, 2025.

The role of genetic and epigenetic factors in sports-related muscle, bone, and brain injuries. Science Progress, 108(4): 1-33, 2025.

The use of dual-energy x-ray absorptiometry (DEXA) in evaluating recovery after musculoskeletal injuries in athletes. The Orthopaedic Journal of Sports Medicine, 13(11): 1-17, 2025.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

12-months MRI after anterior cruciate ligament reconstruction can identify risk factors for subsequent graft rupture and used to guide the return to sport – Results from a high-volume institution. Journal of ISAKOS, 14: 1000994, 2025.

Introduction: Retear after anterior cruciate ligament reconstruction (ACLR) has been reported between 6% and 31% of cases, resulting in worse outcomes and increased risk of post-traumatic osteoarthritis. This study investigated whether postoperative magnetic resonance imaging (MRI) assessment, clinical outcomes, and return-to-sport test findings can help identify patients at higher risk of early graft retear.

Methods: Retrospective analysis of 430 patients who underwent primary ACLR using hamstring autograft between 2017 and 2022, with a minimum follow-up of 12 months. Baseline characteristics, intraoperative, and postoperative information were collected. Graft signal, tunnel widening and positioning were assessed through a 12-month MRI. Patients who experienced early graft retear were compared with a matched cohort with a ratio of 1:4. Continuous variables were compared. Logistic regression and analysis of variance were used to identify risk factors and association with retear. p values < 0.05 were considered significant.

Results: Of the 346 patients who met the inclusion/exclusion criteria, 19 experienced a graft rupture (5.4 %) after the 12-months follow-up. Of those, six patients were excluded due to missing postoperative 1-year MRI. Overall, 13 patients were included in the retear group and compared with 50 matched patients. The mean follow-up was 23.6 ± 4.2 months (range, 18-36 months). The 12-month MRI showed a greater signal-to-noise quotient (SNQ) in case of retear (2.93 vs 2.02; p = 0.029). The analysis of variance showed a positive interaction between graft signal and retear (analysis of variance [ANOVA], p = 0.028). Tunnel positioning was comparable between the groups, and not associated with retear. Similarly, tibial and femoral tunnel widening were comparable between the two groups and not associated with graft retear (ANOVA, p = 0.733 and p = 0.190). A greater proportion of patients had an anterior knee laxity >2 mm at one year in the retear group (83.3% vs 38.8 %, p = 0.058). However, mean laxity was comparable (2.55 mm vs 1.35 mm; p = 0.189). No differences were noted for clinical scores and return-to-sport testing.

Conclusion: Higher graft signal on MRI at 12 months, indicating poorer graft healing, is associated with early anterior cruciate ligament retear. Resonance imaging after ACLR could be used to guide the follow-up management. Further research is required to confirm these preliminary findings.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Changes in eccentric hamstring strength and muscle soreness in younger and older male football players following match play. BMC Sports Science, Medicine and Rehabilitation, 17: 310, 2025.

Introduction: Hamstring strains commonly occur late in football matches. Hamstring injury risk rises with age and fatigue. This study investigated eccentric hamstring strength and perceived hamstring soreness after a football match in younger and older players.

Methods: Maximal eccentric hamstring strength and perceived soreness were assessed in younger (n = 11, 20.2 ± 1.1 years, total playing time: 34.5 ± 8.8 min) and older (n = 10, 27.4 ± 3.7 years, total playing time: 44 ± 10.5 min) professional male football players before and after a friendly match aimed at selecting the primary team for the upcoming season. A two-way mixed model ANOVA was employed for statistical analyses. Furthermore, Hedges' (adjusted) g effect sizes were calculated to evaluate the magnitude of effects.

Results: The older group experienced significant reductions in absolute (-20 N ± 38.5, g = 0.58, p = 0.027) and relative (-0.26 N/kg ± 0.5, g = 0.43, p = 0.038) maximal eccentric hamstring strength, while younger group showed no changes (absolute: 3 N ± 36.1, g = 0.07, p = 0.773; relative: -0.037 N/kg ± 0.54, g = 0.07, p = 0.783). There was no statistically significant difference between groups for the absolute (p = 0.195; g = 0.45) and relative (p = 0.226, g = 0.43) eccentric strength changes. On the other hand, both groups reported increases in perceived hamstring soreness after the match (older: 29% ± 14.1, p < 0.001; younger: 40% ± 14.9, p < 0.001), with the younger group reporting more significantly higher soreness changes than the older group (11% ± 17.2, p = 0.025). A significant negative association existed between age and eccentric strength changes (R = -0.447, R^2 = 0.2, p = 0.007).

Conclusions: The older group showed significant declines in eccentric hamstring strength after the match, with age as a predictor. Specialists should focus on enhancing eccentric strength in older players and monitor their recovery to minimize injury risk. Future psycho-physiological research should examine muscle soreness perceptions and force production in fatigued younger and older football players. The results must be interpreted with caution due to the shorter playing time, unaccounted match workload, and limited sample size. Moreover, future research is necessary, including workload and performance metrics over a longer play duration for each player, with a larger sample size to achieve more reliable conclusions.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Clenbuterol abuse in bodybuilding and athletics: Unsupervised use, psychological motivations, and health consequences. Cureus, 17(5): e84904, 2025.

Abstract - Clenbuterol is a β2-adrenergic agonist originally developed as a bronchodilator but is now often used without supervision for its perceived muscle-building and fat-burning effects. Although it is not approved for human use in many regions, it continues to be used in bodybuilding and athletics for physique and performance enhancement. Its appeal lies in its ability to influence muscle development and fat metabolism. Many individuals use it due to psychological motivators such as body image concerns and a desire for rapid physical changes. However, its use may result in side effects involving the heart, muscles, and overall physical well-being, such as arrhythmias, hypertension, and muscle spasms. In some cases, it has also been linked to accidental exposure through contaminated food. The widespread misuse of clenbuterol raises ethical concerns in competitive environments. More discussion is needed around developing safer methods of achieving similar goals, and there is a need for increased public awareness and caution regarding its use.

Does maximal strength or speed-strength have stronger associations with performance indicators in the rugby football codes? A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-13, 2025.

Background: Both maximal strength and speed-strength are considered key aspects of rugby league and rugby union match-play yet represent distinct physical qualities. Establishing whether maximal strength or speed-strength has a greater association with game performance in these sports can help direct resistance training interventions towards the physical quality most likely to transfer to the outcome of interest. It is therefore important to develop a clearer understanding of whether it is maximal strength or speed-strength that has stronger links to key aspects of rugby union and rugby league competition.

Objectives: To systematically review and meta-analysis the literature to compare maximal strength versus speed-strength measures in their associations with within-game key performance indicators (KPIs) in rugby union and rugby league. Further, sub-group analysis was undertaken to determine

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

whether the type of KPI (tackles, rucks, line breaks, carries/hit-ups, errors, and miscellaneous) moderated the extent to which the observed effect favored maximal strength or speed-strength for a given KPI.

Methods: A systematic search, conducted in accordance with PRISMA guidelines, of Web of Science, PubMed, and SportDiscus was conducted up to December 2024. Studies meeting the inclusion criteria provided correlation coefficients that were converted into Fisher's z-transformed effect sizes to standardize the measures and allow for consistent comparison across studies. These effect sizes were analyzed using a multivariate meta-analysis framework.

Results: Six studies containing 41 maximal strength versus speed-strength comparisons and 134 players were included in the final analysis. Of the 786 studies initially identified, 28 underwent full-text screening, with 22 ultimately excluded. The overall pooled effect was 0.24 [0.11: 0.37] in favor of strength; however, the prediction interval ranged from g = - 0.58 to g = 1.06. The moderator analysis revealed a statistically significant effect of type ($p = 0.038$), indicating stronger associations for maximal strength compared to speed-strength. Residual heterogeneity ($p = 0.005$) suggested substantial variability in effect sizes across studies that was not fully explained by the model.

Conclusions: Within-game rugby union and rugby league KPIs typically have stronger links to maximal strength than to speed-strength. However, the wide prediction interval underscores that the observed maximal strength advantage may not generalize across all types of KPIs or predictably translate to all future settings.

Effect or oral and topical sodium bicarbonate on functional recovery and soccer-specific performance after exercise-induce muscle damage: A randomized, double-blind, placebo-controlled study. Nutrients, 17: 3383, 2025.

Objectives: This study assessed the influence of oral and topical sodium bicarbonate (SB) on recovery and soccer-specific performance after exercise-induced muscle damage (EIMD).

Methods: In a randomized, double-blind, placebo-controlled, parallel group design, 24 soccer players were allocated to oral SB, topical SB (PR Lotion, Momentous), or placebo (PLA) (n = 8 per condition) and

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

attended six laboratory sessions: (i) familiarization, (ii) baseline measures, and (iii) four experimental trials on consecutive days. Muscle damage was induced on day 1 using 40 × 15 m sprints, after which either 0.3 $g \cdot kg^{-1}$ body mass (BM) SB (SB-ORAL), 0.9036 $g \cdot kg^{-1}$ BM PR Lotion (SB-LOTION), or an equivalent PLA was given. Recovery outcomes were measured post-EIMD, 1 d, 2 d, and 3 d post (day 1-4). Soccer-specific performance was repeated 3 d post, with supplements administered again 2 h pre-exercise. Recovery measures included muscle soreness, vertical jump height, and maximal voluntary isometric contraction. Illinois agility test (IAT), 8 × 25 m repeated sprints, and Yo-Yo Intermittent Recovery Test Level 2 (Yo-Yo IR2) were assessed as soccer-specific performance.

Results: Neither SB-ORAL nor SB-LOTION accelerated indices of recovery but decline in soccer-specific performance from baseline to 3 d post was attenuated for SB-ORAL, with significant effects for IAT (p = 0.032, g = 1.69) and Yo-Yo IR2 (p = 0.026, g = 1.61) compared with PLA.

Conclusions: SB did not accelerate recovery following EIMD but prescribing oral SB before subsequent exercise might rescue key performance indicators. These findings offer implications for practitioners working with soccer players during periods where full recovery is not achieved.

Effects of simulated short-haul flights on sport-specific performance, sleep, and psychophysiological responses in football players. Scandinavian Journal of Medicine & Science in Sports, 35: e70169, 2025.

Abstracts - The effects of two short-haul air travel (SHAT) of different durations on sport-specific performance, sleep, and psychophysiological indices in highly trained football players were examined. Nineteen participants completed, in a randomized order, two simulated SHAT, 1 week apart, each lasting 4 h and 2 h, followed by a control condition (CON). Generic (10-m sprint, repeated sprint ability; RSA, countermovement jump; CMJ), sport-specific performance (Hoff Test, Loughborough Soccer Passing Test; LSPT), and psychophysiological responses (sleep, heart rate variability [HRV], salivary cortisol and immunoglobulin-A [sIgA], and perceptual recovery) were evaluated before (Baseline) and after each condition. Compared to Baseline, RSA and 10-m sprint declined by 3.5% and 6%, respectively, on the day after travel (D + 1) ($p < 0.001$ and $p < 0.01$). Likewise, CMJ and total distance covered during the Hoff test decreased by 5.5% at D + 1 ($p < 0.001$). LSPT performance decreased only after the 4 h

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

flight ($p < 0.05$). Total sleep time was reduced the night before SHAT ($p < 0.001$) compared to Baseline. sIgA was reduced on the day of the travel (DT) and D + 1 after both SHAT, while HRV was decreased at DT only after the 4 h SHAT ($p < 0.05$). Mood states remained unaffected by travel, regardless of duration. We currently suggest that short-haul air travel, whether 4 or 2 h, induces psychophysiological perturbations and performance deterioration, likely due to the travel restrictive conditions and the inadequate sleep obtained the night before the flight. Since both flights were conducted early in the morning, more research is warranted to investigate the role of departure time on football performance.

Ensuring scientific rigor in research: Why sports-related publicly obtained data fall short. Foot & Ankle Orthopaedics, 10(2): 1, 2025. (No abstract available)

Exploring the physiological mechanisms of sleep's influence on athletic performance and recovery: A narrative review. Sleep and Breathing, 29: 354, 2025.

Purpose: This narrative review explores the complex relationship between sleep, athletic performance, and recovery, emphasizing the physiological mechanisms at play.

Methods: This review included papers reporting physiological mechanisms linking sleep, athletic performance, and recovery. Literature was identified through targeted searches in relevant databases and bibliographies of referenced scientific articles. Retrieved studies were analyzed thematically according to major domains, enabling integration of foundational mechanisms with applied evidence to inform practice and future research.

Results: Slow-wave sleep (SWS) plays an important role in physical recovery, with its increase during the night being linked to rises in metabolic stress and growth hormone production. When training volume and/or intensity increase, a corresponding increase in SWS is usually observed, but a fine line may exist between promotion and disruption of sleep (i.e., overtraining). Acute sleep deprivation or restriction may influence sports performance differently, depending on the sport. In sports requiring complex skills

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

or high cognitive demands, impairments may be more pronounced. Among the main strategies, napping has received the greatest attention, as it serves as a supplement to nighttime sleep and is associated with improvements in fatigue and sleepiness perception and physical and cognitive performance. Sleep extension has been studied less, but existing evidence shows improvements in cognitive and physical performance, as well as sport-specific skill execution.

Conclusions: Despite the growing body of research on athletes' sleep patterns and their impact on performance, there is limited focus on the underlying mechanisms and practical benefits of sleep.

Global research trends in sports nutrition and football over the last 20 years (2004-2024). Sports, 13: 365, 2025.

Background: We aimed to map the scientific production on sports nutrition applied to soccer.

Methods: A scientometric analysis was performed using articles published between 2004 and 2024, retrieved from Web of Science, PubMed, and Scopus. The search yielded 2636 documents, and 526 original articles were included after removing reviews, meta-analyses, duplicates, and studies outside the scope. Data were analyzed using Bibliometrix version 5.0.1; Massimo Aria & Corrado Cuccurullo; Naples; Italy. and VOSviewer version 1.6.20; Centre for Science and Technology Studies (CWTS), Leiden University; Leiden; The Netherlands software.

Results: There was a 1.450% increase in publications over the period, with a peak in 2024. *Nutrients* was the leading publication source, while Morton J. and Maughan R. were the most productive authors. Liverpool John Moores University stood out as a collaboration hub. The United Kingdom 371 took the lead in both publication volume and citations. Early research trends focused on hydration and dietary optimization, whereas recent studies emphasized low energy availability, polyphenols, anthropometry, and recovery strategies. The conceptual structure focused on terms such as sports, nutrition, energy intake, food intake, performance, soccer, and training load. Peripheral terms included fluid balance and sweat rate. The co-occurrence analysis revealed underexplored topics such as oxidative stress, lipid peroxidation, beta-alanine supplementation, and antioxidant markers.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Conclusions: Advancing these research areas is essential to consolidating nutritional strategies with direct effects on performance and health in soccer players.

Gluteal muscle forces during hip-focused injury prevention and rehabilitation exercises. Medicine and Science in Sports and Exercise, 55(4): 650-660, 2025.

Purpose: This study aimed to compare and rank gluteal muscle forces in eight hip-focused exercises performed with and without external resistance and describe the underlying fiber lengths, velocities, and muscle activations.

Methods: Motion capture, ground reaction forces, and electromyography (EMG) were used as input to an EMG-informed neuromusculoskeletal model to estimate gluteus maximus, medius, and minimus muscle forces. Participants were 14 female footballers (18-32 yr old) with at least 3 months of lower limb strength training experience. Each participant performed eight hip-focused exercises (single-leg squat, split squat, single-leg Romanian deadlift [RDL], single-leg hip thrust, banded side step, hip hike, side plank, and side-lying leg raise) with and without 12 repetition maximum (RM) resistance. For each muscle, exercises were ranked by peak muscle force, and k-means clustering separated exercises into four tiers.

Results: The tier 1 exercises for gluteus maximus were loaded split squat (95% confidence interval [CI] = 495-688 N), loaded single-leg RDL (95% CI = 500-655 N), and loaded single-leg hip thrust (95% CI = 505-640 N). The tier 1 exercises for gluteus medius were body weight side plank (95% CI = 338-483 N), loaded single-leg squat (95% CI = 278-422 N), and loaded single-leg RDL (95% CI = 283-405 N). The tier 1 exercises for gluteus minimus were loaded single-leg RDL (95% CI = 267-389 N) and body weight side plank (95% CI = 272-382 N). Peak gluteal muscle forces increased by 28-150 N when exercises were performed with 12RM external resistance compared with body weight only. Peak muscle force coincided with maximum fiber length for most exercises.

Conclusions: Gluteal muscle forces were exercise specific, and peak muscle forces increased by varying amounts when adding a 12RM external resistance. These findings may inform exercise selection by

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

facilitating the targeting of individual gluteal muscles and optimization of mechanical loads to match performance, injury prevention, or rehabilitation training goals.

Letter to the Editor: Injectable therapeutic peptides – An adjunct to regenerative medicine and sports performance? Arthroscopy: The Journal of Arthroscopic and Related Surgery, 41(2): 150-151, 2025.

Abstract - High-level athletes and bodybuilders are constantly seeking novel therapies to enhance recovery and expedite return from injury-injectable peptides are a new and trending therapy that may be the wave of the future in the realm of regenerative medicine research in treating joint injuries and osteoarthritis. Very early in vivo research on pharmacokinetics indicates the possibility that body protection compound 157 (BPC-157) is at the forefront of therapeutic peptides, with early demonstrations of this experimental peptide optimizing endurance training, metabolism, recovery, and tissue repair. Although unregulated and yet readily available for purchase over the internet, there is scarce orthopaedic literature investigating the clinical use and outcomes of such therapeutic peptides in tendon, muscle, and cartilage injury. However, this has not slowed the recent exponential growth of the multi-billion-dollar industry in the development of therapeutic peptides. As orthopaedic surgeons and team physicians, we should stay up to date with the latest pharmacokinetic, safety, ethical, and legal profiles and regulations regarding synthetic peptide supplementation for injury recovery and sports performance optimization in our patients, from elite athletes to fitness fanatics, because they will continue to seek the latest and greatest in treatment options and will be approaching us with questions on their results, risks, and benefits.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Intermediary role of mental toughness beliefs on the relationship between pain self-efficacy and fear avoidance in elite injured athletes. BMC Sports Science, Medicine and Rehabilitation, 17: 124, 2025.

Introduction: Athletes adopt different coping mechanisms with injury. Differences and links between fear avoidance (FA), pain self-efficacy (PSE), and mental toughness (MT) might affect injury outcomes.

Aim: To examine the relationship between Fear Avoidance, Pain Self-Efficacy and Mental Toughness in injured elite and competitive athletes in Lebanon.

Methods: This is a cross-sectional study that included 172 athletes. Different questionnaires were used to determine the three pre-mentioned concepts. FA, PSE, and MT were measured using AFAQ-Athletic Fear Avoidance Questionnaire, PSEQ-Pain Self-Efficacy Questionnaire, and SMTQ-Sports Mental Toughness Questionnaire respectively. The data was collected and analyzed using IBM SPSS software.

Results: Athletes who practiced boxing reported the highest SMTQ score. Age and training hours predicted higher scores in mental toughness while athletes returning to practice within the last month showed higher PSEQ scores. A negative relationship was found between SMTQ and AFAQ ($r = -0.47$, $p < 0.001$) as well as between PSEQ and AFAQ ($r = -0.44$, $p < 0.001$). However, a positive relationship was seen between SMTQ and PSEQ ($r = 0.36$, $p < 0.001$).

Conclusion: By incorporating insight into FA, PSE, and MT, sports professionals may enhance their ability to assess both physical and psychological predicaments of injured athletes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Melatonin secretion and impacts of training and match schedules on sleep quality, recovery, and circadian rhythms in young professional football players. Biomolecules, 15: 700, 2025.

Abstract - Modern elite football is becoming increasingly physically demanding, often requiring training and matches to be played at night. This schedule may disrupt circadian rhythms and melatonin secretion, thereby impairing sleep and recovery. This study investigated the effects of training time on melatonin secretion, circadian phase markers, and sleep parameters in elite youth soccer players. Forty male players (aged 16–18 years) from an elite Spanish youth football club were studied. Two groups followed the same training program but trained either in the morning (MT) or in the evening (ET). Salivary melatonin was measured at six time points to determine the mean levels, dim light melatonin onset (DLMO), amplitude, and acrophase. Chronotype, sleep quality (PSQI), and daytime sleepiness (ESS) were assessed using validated questionnaires. Dietary intake and anthropometric variables were also recorded. The MT group had higher mean melatonin levels ($p = 0.026$) and earlier DLMO ($p = 0.023$) compared to the ET group. Sleep quality was significantly better in the MT group ($p < 0.001$), despite shorter sleep duration ($p = 0.014$). No major differences in diet or anthropometry were observed. The chronotype had a secondary effect on the circadian markers. Evening training is associated with alterations in melatonin rhythms and reduced sleep quality, possibly due to light-induced chronodisruption. These findings highlight the importance of training timing as a modifiable factor in the chronobiology and recovery of athletes. Incorporating circadian principles into training schedules may optimize resting time and thus performance and long-term health in athletes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Neither American football playing position nor total years playing football are associated with neurobehavioral health outcomes: An NFL-LONG study. Current Sports Medicine Reports, 24(11): 372-380, 2025.

Abstract - Years playing American football and primary playing position are often used to estimate concussion exposure. We examined associations between playing position and total years played with concussion history and neurobehavioral outcomes among former (N = 1544) National Football League players in the NFL-LONG study. Outcomes included the PROMIS Cognitive, Depression, and Anxiety T-scores, and Neuro-QOL Emotional-Behavioral Dyscontrol T-scores. Playing position was categorized as offensive/defensive linemen, running back/linebacker/tight end, quarterback/wide receiver/defensive back, and special teams. Analysis of variance tested differences between years played across position groups. Multivariable regressions estimated associations between years played, position group, and their interaction for outcomes, adjusting for covariates. There was no significant difference in years played across playing positions, and no statistically or clinically significant findings linking position, years played, or their interaction with negative neurobehavioral outcomes. These findings suggest concussion history should be directly assessed and considered alongside other exposure indicators to better understand neurobehavioral outcomes.

Physiological adaptations and practical efficacy of different blood flow restriction resistance training modes in athletic populations. Frontiers in Physiology, 16: 1683442, 2025.

Abstract - Blood flow restriction resistance training enhances athletic adaptations via distinct mechano-metabolic pathways. This review synthesizes evidence comparing three blood flow restriction resistance training modalities: Low-load resistance training with blood flow restriction (using 20%–30% of one-repetition maximum) prioritizes metabolic stress (lactate and hydrogen ion accumulation, cellular swelling), activating growth hormone (GH)/insulin-like growth factor 1 (IGF-1)/mechanistic target of rapamycin (mTOR) pathways to promote type I muscle fiber hypertrophy, making it suitable for joint-sparing rehabilitation scenarios. Supplemental blood flow restriction resistance training programs

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

combine high-load tension (utilizing 75%–90% of one-repetition maximum) with additional blood flow restriction to produce an acute synergistic effect. This method enhances the recruitment of type IIa/x muscle fibers and prolongs mTOR phosphorylation. Combined blood flow restriction resistance training employs alternating cycles of high-load phases (70%–85% 1RM) and blood flow restriction phases (hypoxia-inducible factor 1-alpha (HIF-1α)-mediated angiogenesis), optimizing phosphocreatine resynthesis and neural drive to achieve specialization of type IIx muscle fibers. Periodized application requires matching modalities with training phases: combined blood flow restriction training for hypertrophy during the preparatory phase, supplemental blood flow restriction training for strength maintenance during the competitive phase, and low-load resistance training with blood flow restriction for active recovery. This mechanistic framework supports evidence-based blood flow restriction resistance training programming to maximize athletic adaptations while mitigating injury risk.

Reciprocal relationships between sleep quality, mental health and the quality of life in elite athletes: A pilot study. Biology of Sport, 43: 267-279, 2026.

Abstract - We aim to investigate the relationship between sleep quality, psychological health, and quality of life (QOL) in highly trained athletes. Elite athletes (n = 118, 20.1 ± 0.64 years; 39 females; 50 world class; 102 aged ≤ 25 years; and 76 practicing individual sports) responded to the Pittsburgh sleep quality index (PSQI), insomnia severity index (ISI), Epworth sleepiness scale (ESS), Depression, anxiety and stress scale (DASS), world health organization QOL (WHOQOL), and bespoke questions related to sleep hygiene. High percentages of the sample reported low or very low sleep quality (62%), moderate or excessive daytime sleepiness (51%), sleeping 7 hours or less (60%), and moderate or severe insomnia (16%), implying a modest sleep health. Female athletes reported higher PSQI ($p < 0.05$; $d = 0.25$), ESS ($p < 0.05$; $d = 0.37$) and DASS ($p < 0.05$; $d = 0.27$) scores compared to males. Young athletes (i.e., ≤ 25 years) reported higher PSQI ($p < 0.05$; $d = 0.49$) and DASS ($p < 0.05$; $d = 0.34$) scores compared to older athletes (i.e., > 25 years). Individual-sport athletes reported higher ESS ($p < 0.05$; $d = 0.37$) and lower QOL ($p < 0.01$; $d = 0.51$) scores compared to team-sport athletes. Higher DASS scores were associated with higher PSQI ($t = 3.68$; $\beta = 0.3$) and ISI ($t = 4.78$; $\beta = 0.36$) scores. Lower physical health (i.e., sub-scale of WHOQOL) was associated with higher DASS ($t = -5.01$; $\beta = -0.42$) and ISI ($t = -8.02$; $\beta = -0.61$) scores. Higher PSQI scores contributed to lower WHOQOL scores ($t = -4.81$; $\beta = -0.41$). In summary, the current

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

study highlights reciprocal relationships between low sleep quality, low mental health and low QOL. Elite athletes (especially sub-groups of female, individual, and young athletes) showed a low sleep quality, potentially affecting their physical and psychological health and QOL.

Should GPS data be normalized for performance and fatigue monitoring in soccer? A theoretical-practical discussion on high-speed running. Frontiers in Sports and Active Living, 7: 1603767, 2025.

Abstract - High-speed running (HSR) is one of the performance metrics of interest, as the volume of HSR during matches has been increasing over the last decade, which suggests that weekly training loads should be adjusted to align with this trend, enabling players to cope with match demands. However, the use of HSR thresholds lacks a solid rationale for their application and fails to account for individual player capacities, likely not reflecting their actual HSR efforts. As such, this theoretical-discussion provides important implications for training prescription, aiming to optimize performance and minimize fatigue. It emphasizes the significant differences in the conceptualization of HSR and highlights the advantages of adopting a normalized approach that reflects the physiological, mechanical and neuromuscular aspects related to HSR, as well as the intermittent profile of football matches. Practical HSR threshold definitions tailored to the capacities of each athlete are proposed, enabling a more evidence-based approach for the interpretation of training loads and game/player profiling. More specifically, within our proposal, HSR can be subdivided into two types: (1) HSR-1, characterized by an entry threshold based on a normalized critical speed, and (2) HSR-2, defined by an entry threshold corresponding to 75% of the athlete's maximum speed.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Specificity and areas of usage of cardiovascular prediction models among athletes – State of the art review. Reviews in Cardiovascular Medicine, 26(5): 37493, 2025.

Abstract - Cardiovascular diseases are a leading cause of mortality worldwide. Physical activity is linked with a reduced prevalence of cardiovascular diseases. However, excessive over-volume of training could negatively increase the risk of cardiovascular diseases. Prediction models are usually derived to facilitate decision-making and may be used to precisely adjust the intensity of physical activity and stratify individual exercise capacity. Incorporating prediction models and knowledge of risk factors of cardiovascular diseases allows for the accurate determination of risk groups among athletes. Due to the growing popularity of amateur physical activity, as well as the high demands for professional athletes, taking care of their health and providing precise pre-participation recommendations, return-to-play guidelines or training intensity is a significant challenge for physicians and fitness practitioners. Athletes with confirmed or suspected cardiovascular disease should be guided to perform training in carefully adjusted safe zones. Indirect prediction algorithms are feasible and easy-to-apply methods of individual cardiovascular disease risk estimation. Current knowledge about the usage of clinical forecasting scores among athletic cohorts is limited and numerous controversies emerged. The purpose of this review is to summarize the practical applications of the most common prediction models for maximal oxygen uptake, cardiac arrhythmias, hypertension, atherosclerosis, and cardiomyopathies among athletes. We primarily focused on endurance disciplines with additional insight into strength training. The secondary aim was to discuss their relationships in the context of the clinical management of athletes and highlights key understudied areas for future research.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Speeding up or slowing down: The effect of decoupling speed on knee biomechanics and limb loading during split-belt treadmill training in persons with ACL reconstruction. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2025.

Background: Surgical limb underloading is a common biomechanical adaptation after anterior cruciate ligament reconstruction (ACLR) and has been linked to early degenerative changes in knee cartilage, which are considered precursors to posttraumatic osteoarthritis. Split-belt treadmill training is an emerging rehabilitation approach that modifies load through asymmetric gait behavior, in which one limb walks faster than the other. While previous research has indicated that split-belt treadmill training can modify limb loading, its effects on post-ACLR biomechanics remain unexplored.

Purpose/hypothesis: The purpose of this study was to examine the effects of decoupling speed on knee biomechanics and limb loading and model their relationship. It was hypothesized that at faster decoupling speeds, knee loads would increase, and at slower speeds, they would decrease.

Study design: Controlled laboratory study.

Methods: Knee joint biomechanics were evaluated while 24 participants (15 females; mean age, 23.5 ± 6.5 years; mean height, 1.72 ± 0.08 m; mean mass, 75.61 ± 13.83 kg; mean postoperative time, 7.94 ± 1.74 months) with ACLR walked on an instrumented treadmill (2000 Hz) synced with a 12-camera motion capture system (200 Hz). Participants completed 5 minutes of baseline walking at 1.1 m/s on tied treadmill belts. Afterward, the authors manipulated the speed of the belt under the ACL-reconstructed leg with 8 randomized 5-minute decoupled speed configurations (ACL-reconstructed limb at 30%-170% of 1.1 m/s in 20% increments) with 2 minutes of tied-belt walking between each speed. Bilateral sagittal plane knee moments, angles, and vertical ground-reaction forces were calculated. Statistical parametric mapping was used to evaluate the effects of decoupling speed on the dependent variables during the stance phase of the gait.

Results: A clear dose-response relationship between decoupling speed and knee/limb loading was found. Specifically, faster decoupling speeds (130%-170% of 1.1m/s) generally increased peak loads and slower speeds (30% and 50% of 1.1 m/s) decreased peak loads during early stance compared with tied-belt walking for both limbs. In contrast, slower decoupling speeds (30%-90% of 1.1 m/s) led to higher

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

bilateral knee/limb loads at midstance, while faster decoupling speeds (130%-170% of 1.1 m/s) resulted in less knee/limb loading during midstance.

Conclusion: These findings suggest that split-belt treadmill training offers a promising method for modulating knee/limb loading post-ACLR.

Clinical relevance: Split-belt treadmill training could be a viable intervention to target the loading asymmetry that is prevalent in persons post-ACLR.

Terminology and interpretation across neuromuscular profiling methods: A systematic review. Sports Medicine, Ahead of Print: 1-31, 2025.

Background: Neuromuscular profiling (NP) provides an objective means of assessing neuromuscular function to guide targeted training interventions. However, coaching terminology used to describe NP outputs (e.g., 'explosive', 'springy') is often inconsistently defined or applied, leading to misinterpretation and limiting effective use. Variations in how NP metrics are calculated and interpreted further complicate comparisons across studies and practical applications. Despite their widespread use in sport science and performance settings, no framework currently exists to evaluate whether the terminology and interpretations associated with NP metrics accurately represent measured outputs and intended neuromuscular traits.

Objectives: This review aimed to evaluate whether terms/labels used across six commonly applied NP methods-Dynamic Strength Index, Eccentric Utilization Ratio, Force-Velocity Profile, Impulse Analysis, Reactive Strength Index, and Reactive Strength Index Modified-accurately represent the neuromuscular characteristics they are intended to measure. A secondary objective was to identify methodological limitations that may affect the interpretation, standardization, and practical application of these methods.

Methods: A systematic search of PubMed, Scopus and SPORTDiscus identified 1653 eligible studies, of which 184 met inclusion criteria. A modified quality scale was used to assess study relevance. Extracted data included reported output values, assigned terms/labels, underlying mechanisms, and supporting evidence. Results were grouped by explanatory themes, with terms/labels identified and examined

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

based on the underlying mechanisms they were intended to represent. Findings were then synthesized to identify patterns in terminology, interpretation, and limitations.

Results: A total of 209 NP methods were evaluated (147 ratio-based, 62 non-ratio-based). NP interpretation was task-dependent and influenced by methodological variability. Inconsistency was found in the terminology, definitions, and calculations used to derive NP metrics. Several studies attempted to classify athletes based on NP metrics, though few accounted for constituent components in interpretation. Commonly used terms/labels included: 'explosive/explosiveness' (indicating greater rate of force development), 'spring/springy' (suggesting effective stretch-shortening cycle utilisation), 'stiff/compliant' (representing a continuum of lower extremity stiffness), and 'strong/weak' (reflecting maximal force production capability).

Conclusions: This review offers a consolidated synthesis of NP terminology, interpretation, and methodological considerations. Clarifying terminology and addressing limitations may improve communication between researchers and practitioners, supporting more standardized use of NP tools.

The association between central nervous system function and core body temperature in patients with exertional heat stroke. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2025.

Objective: Describe central nervous system (CNS) function using the Glasgow Coma Scale (GCS) and a description of signs and symptoms in patients with exertional heat stroke (EHS) during the course of treatment at a warm-weather road race. Determine whether a relationship exists between rectal temperature (Tre) and the GCS in patients with EHS during treatment.

Design: Cross-sectional research design retrospectively using medical charts.

Setting: Eleven-kilometer road race medical tent.

Patients: Patients (n = 25) with EHS receiving treatment.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Interventions: Dedicated scribes recorded patient vitals on a medical chart. The deidentified medical charts were provided by the medical director after the race. A Spearman rank correlation was used to determine whether a relationship existed between Tre and GCS (subscales and cumulative).

Main outcome measures: Rectal temperature was measured using an in-dwelling flexible probe with temperatures recorded approximately every 3 minutes. GCS was recorded by subscales on the medical chart by scribes according to existing word anchors.

Results: Average initial Tre was 41.0 ± 0.6°C (105.8 ± 1.1°F) and cumulative GCS score was 14 ± 3 (4-15). No single CNS dysfunction predominated across most observations. Confusion had the highest recorded percentage (19.8%). There was a significant but weak correlation between Tre and GCS [rs(124) = -0.255, P = 0.004; P2 = 0.06]. For time points in which Tre was >40°C, there was no correlation between Tre and GCS [rs(60) = -0.209, P = 0.108; P2 = 0.04], and GCS subscales Eye [rs(60) = -0.230, P = 0.077; P2 = 0.05], Verbal [rs(60) = -0.183, P = 0.161; P2 = 0.03], and Motor [rs(60) = -0.214, P = 0.100; P2 = 0.05].

Conclusions: Although GCS is a good global measure, it may not be appropriate in capturing how CNS dysfunction presents in patients with EHS. Clinicians should continue to use clinical CNS dysfunction and Tre to initially recognize EHS, however, Tre should be used to guide prehospital cooling of patients with EHS.

The impact of weekly acceleration and deceleration loads on neuromuscular performance in soccer: A session-to-session analysis. Journal of Sports Science and Medicine, 24: 851-860, 2025.

Abstract -This study examined weekly variations in accelerations, decelerations, and neuromuscular performance in male under-19 soccer players. It also explored the relationship between accumulated acceleration and deceleration loads and fluctuations in neuromuscular performance across a weekly microcycle. A repeated-measures observational design was used over two consecutive weeks, involving 39 outfield players monitored via Global Navigation Satellite Systems. Neuromuscular performance was assessed through countermovement jump (CMJ), hamstring strength (HS), and delayed onset muscle soreness (DOMS), measured both before and after training on five days each week, following a

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

consistent training structure. Significant post-training reductions in CMJ (all days $p < 0.001$) and HS (all days $p < 0.001$) were observed, with the greatest impairments occurring on MD-4 and MD-5, although CMJ showed a slight rebound on MD-5 compared to MD-4. DOMS ratings peaked on MD-3 ($F = 39.186$, $p < 0.001$), indicating a buildup of midweek fatigue. The highest numbers of accelerations ($F = 248.121$, $p < 0.001$) and decelerations ($F = 227.853$, $p < 0.001$) occurred on MD-4 and MD-5. However, no statistically significant correlations were found between daily acceleration/deceleration counts and changes in CMJ or HS performance. These findings suggest that while neuromuscular performance declines progressively throughout the week, no statistically significant correlations were found between impairments and the acceleration or deceleration load of individual training sessions. Instead, the cumulative effect and timing of high-intensity training may play a more substantial role in contributing to fatigue and neuromuscular decline.

The role of genetic and epigenetic factors in sports-related muscle, bone, and brain injuries. Science Progress, 108(4): 1-33, 2025.

Abstract - Understanding how genetic and epigenetic factors influence predisposition, susceptibility, protection, severity, and recovery potential in sports-related muscle, bone, and brain injuries is gaining increasing attention, as these injuries represent a major concern for athletes' health and performance. This narrative review explores the role of specific genetic variants, such as those in *ACE, ACTN3, AMPD1, VDR,* and *APOE*, in predisposing athletes to injury. In parallel, the emerging role of microRNAs, including miR-21, miR-133a, miR-208a, and miR-214 in regulating gene expression related to muscle regeneration, inflammation, and neuroprotection, is highlighted. These small noncoding RNAs are influenced by environmental stimuli such as sports activity and training, and may serve as promising biomarkers for injury risk and recovery. This review is based on a narrative approach, involving targeted literature searches in databases such as PubMed, Scopus, and Web of Science, with no restrictions on language or publication date. Articles were selected based on relevance using predefined keywords and were categorized by injury type (muscular, bone, and concussion-related). However, several limitations remain, including population heterogeneity, small sample sizes, and the lack of functional validation of identified variants, which currently limits their clinical application. Ethical considerations surrounding the use of genetic data in athletes also need to be addressed. In the future, better integration of

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

multiomic data could support the development of personalized and ethically sound strategies for injury prevention and athlete care.

The use of dual-energy x-ray absorptiometry (DEXA) in evaluating recovery after musculoskeletal injuries in athletes. The Orthopaedic Journal of Sports Medicine, 13(11): 1-17, 2025.

Background: Dual-energy x-ray absorptiometry (DEXA) is a widely used imaging modality in sports medicine and orthopaedics due to its accuracy in assessing bone mineral density (BMD) and body composition. Tracking these parameters provides valuable insights into the recovery process after musculoskeletal injuries, which are prevalent among athletes.

Purpose: To evaluate how DEXA imaging has been used to guide rehabilitation in both recreational and competitive athletes by monitoring changes in BMD and body composition after musculoskeletal injuries.

Study design: Scoping review; Level of evidence, 4.

Methods: A comprehensive literature search was conducted in Ovid MEDLINE, EMBASE, Cochrane Central, and SportDISCUS databases (2000-2024). Studies were included if they used DEXA to measure BMD or body composition changes in athletes after musculoskeletal injuries. Titles, abstracts, and full texts were screened independently by 2 reviewers, with discrepancies resolved through discussion. Data were synthesized qualitatively, and major trends were reported and identified.

Results: Of 1132 unique records, 12 studies met inclusion criteria, involving 319 athletes (34% female) with injuries such as anterior cruciate ligament (ACL) tears, lumbar stress fractures, femoroacetabular impingement, and Achilles tendon ruptures. Athletes represented a range of sports, including soccer, basketball, cricket, and triathlon, at both competitive and recreational levels. DEXA identified significant BMD declines after injury, particularly in the affected limb. For instance, up to a 7% BMD reduction was observed in surgical limbs after ACL reconstruction, persisting for up to 2 years in some cases. Rehabilitation strategies incorporating blood flow restriction therapy or combined running and isometric

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

exercises better preserved lean mass and BMD compared with conventional programs focused solely on closed and open kinetic chain exercises.

Conclusion: The review demonstrated that DEXA imaging is a promising tool in orthopaedic sports medicine for assessing injury-related changes in BMD and body composition. It provides detailed insights into recovery processes and aids in tailoring rehabilitation strategies and return-to-sport decisions. Some studies incorporated advanced rehabilitation methods such as blood flow restriction therapy, which appeared to accelerate recovery as measured by DEXA outcomes. Future studies should explore integrating DEXA findings into athlete-specific rehabilitation protocols with validated thresholds for return to play.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/